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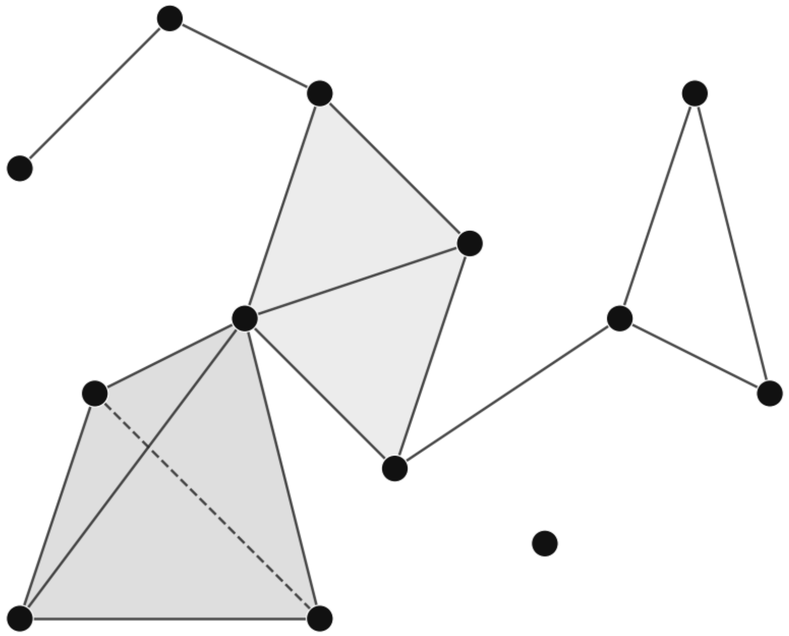
Lecture 6: Persistent Homology

Geometry and Topology in Machine Learning

Inés García Redondo — October 20, 2026

Recap: simplicial complexes

Definition



Def. (Abstract) Simplicial Complex

Given a discrete set $S = \{s_1, \ldots, s_n\}$, a simplicial complex X is a set of subsets $\sigma \subset S$ *closed under inclusion* (if $\sigma \in X$, then for all $\tau \subseteq \sigma$ we must have $\tau \in X$)

We can *stratify* a simplicial complex by dimension:

Vertices: 0-dimensional simplices

Edges: 1-dimensional simplices

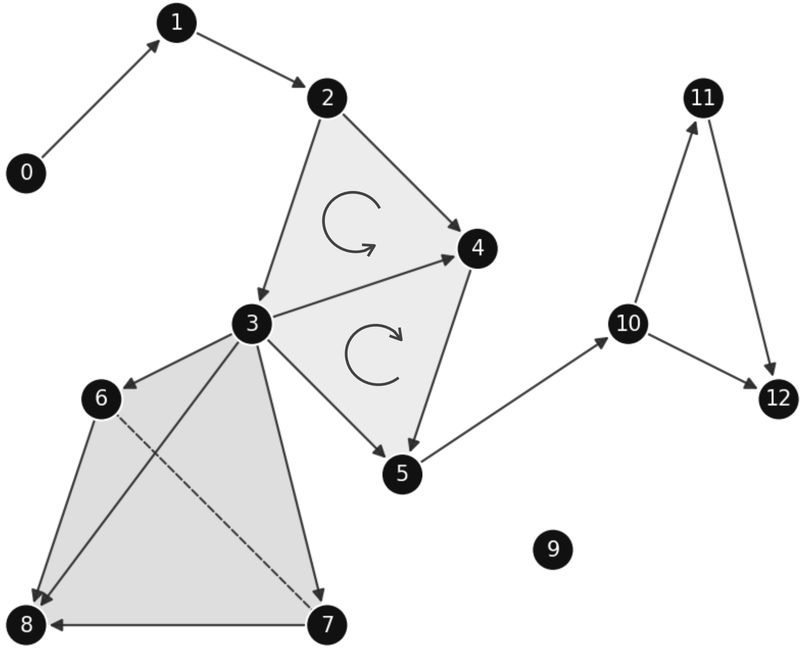
Triangles: 2-dimensional simplices

In general:

- k-**dimensional simplex**: $\sigma = \{s_{i_0}, \ldots, s_{i_k}\}$ $k+1$ elements!
- X_k: set of k-dimensional simplices
- k-**skeleton** $(X_{\leq k})$: all simplices with dimension $\leq k$

Recap: simplicial homology

Orientations



The first thing we do is choosing an **orientation** for the simplices, that is, a unique way of representing a set $\sigma = \{s_{i_0}, \ldots, s_{i_k}\}$

Note that we use the same notation that we had in graphs to distinguish oriented and un-oriented simplices!

How to choose an orientation?

- Choose an ordering of the vertices $S = \{1,\ldots,n\}$
- Orient higher order simplices accordingly: $(s_{i_0}, \ldots, s_{i_k})$, $i_0 < \ldots < i_k$
- We will simply denote the simplices as: $(s_{i_0}, \ldots, s_{i_k}) \equiv (i_0, \ldots, i_k)$

- **Any other ordering of the vertices is related to this one by**:

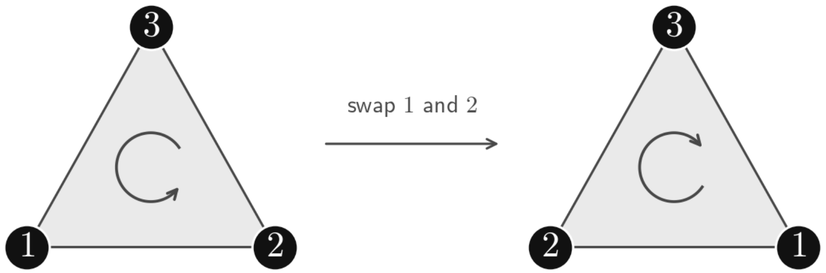
$$(i_0, \ldots, i_k) = (-1)^{\mathfrak{p}}(j_0, \ldots, j_k)$$

Where $\{s_{i_0}, \ldots, s_{i_k}\} = \{s_{j_0}, \ldots, s_{j_k}\}$ (the simplices contain the same vertices) and $\mathfrak{p}$ is the number of pairwise permutations to go from $(s_{j_0}, \ldots, s_{j_k})$ to $(s_{i_0}, \ldots, s_{i_k})$

Examples:

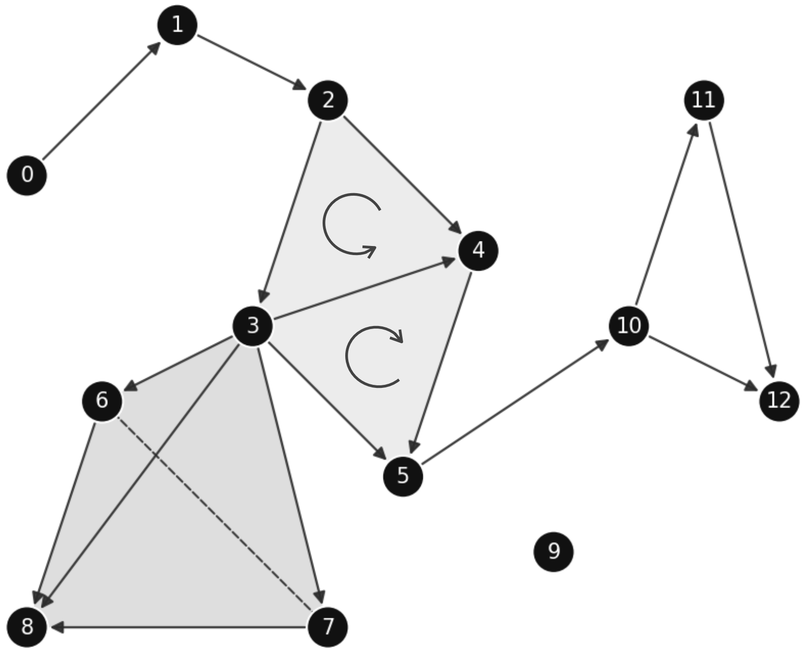
- $(1,2) = -(2,1)$
- $(1,2,3) = -(2,1,3) = (2,3,1) = -(3,2,1)$

$(1,2,3) \quad = \quad -(2,1,3)$



Recap: simplicial homology

Chain spaces



Chain space: let $k \geq 0$, the k-chain space is the *vector space* spanned by the *oriented simplices*

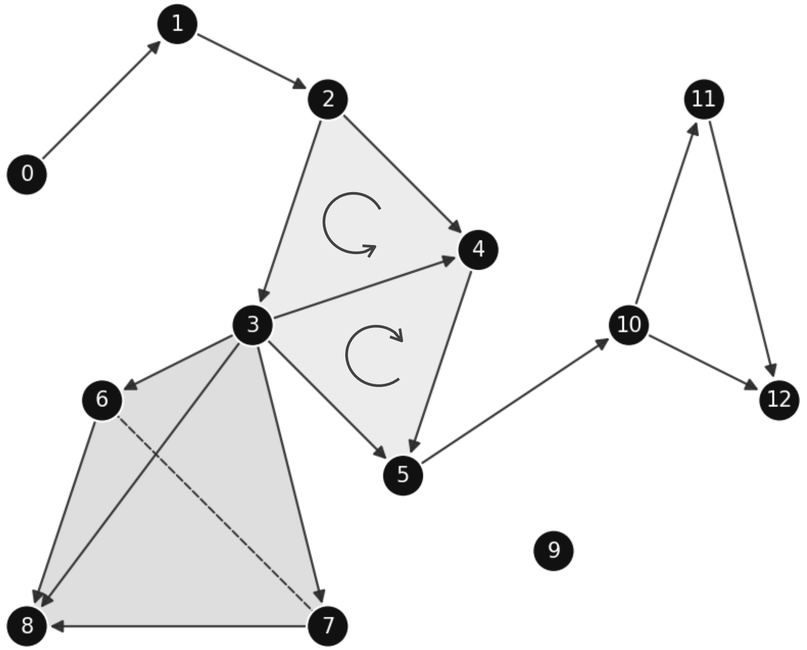
$$C_k(X) = \text{span}\{(i_0, \ldots, i_k) : i_0 < \cdots < i_k\} \cong \mathbb{R}^{|X_k|}$$

$$c = \sum_{\sigma \in X_k} a_\sigma \cdot \sigma, \quad a_\sigma \in \mathbb{R}$$

Try it: What are the chain groups of the complex on the left?

Recap: simplicial homology

Boundary operators



Boundary operator: *linear map* connecting chain spaces of consecutive dimension $\partial_k : C_k(X) \to C_{k-1}(X)$ defined on the basis elements as

$$\partial_k(i_0, \ldots, i_k) = \sum_{j=0}^{k} (-1)^j \ (i_0, \ldots, \widehat{i_j}, \ldots i_k)$$

And extended linearly:

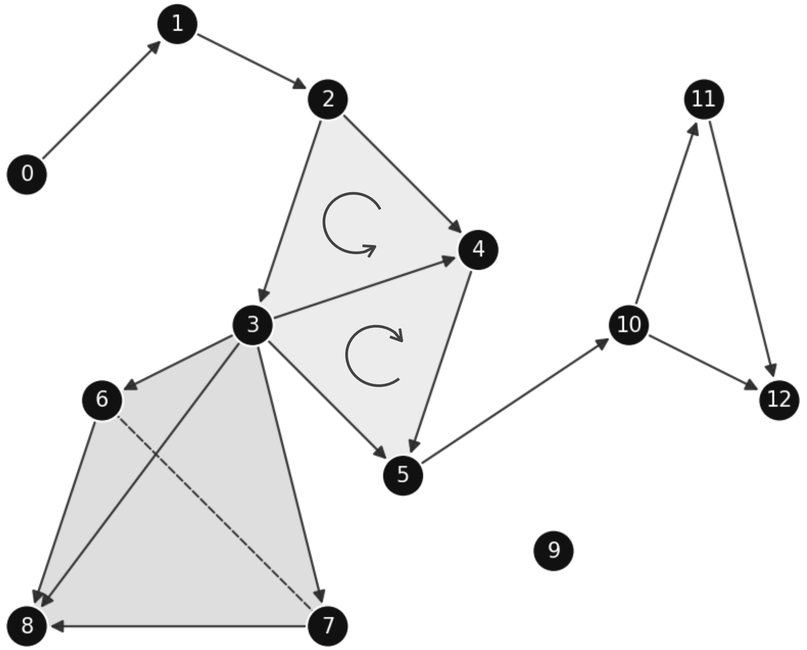
$$c = \sum_{\sigma \in X_k} a_\sigma \cdot \sigma, \quad a_\sigma \in \mathbb{R} \qquad\qquad \partial_k c = \sum_{\sigma \in X_k} a_\sigma \cdot \partial_k \sigma, \quad a_\sigma \in \mathbb{R}$$

Try it:

$$\partial_1(0,1) =$$
$$\partial_2(2,3,4) =$$
$$\partial_3(3,6,7,8) =$$

Recap: simplicial homology

Boundary operators



Boundary operator: *linear map* connecting chain spaces of consecutive dimension $\partial_k : C_k(X) \to C_{k-1}(X)$ defined on the basis elements as

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Try it:

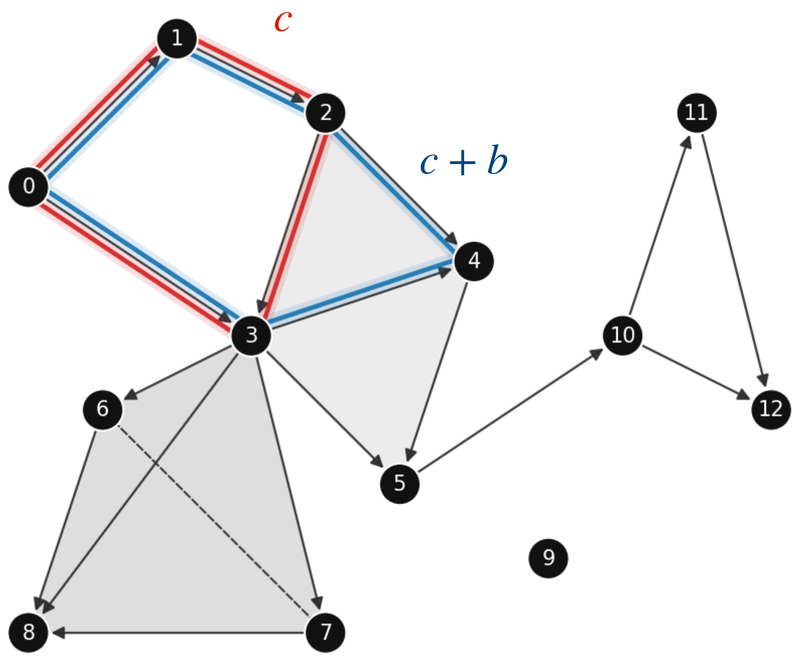
$$\partial_1(0,1) =$$
$$\partial_2(2,3,4) =$$
$$\partial_3(3,6,7,8) =$$

Important property:

$$\partial_k \, \partial_{k+1} = 0 \implies \ker \partial_k \subseteq \operatorname{im} \partial_{k+1}$$

Recap: simplicial homology

Homology groups



If I can move through filled in triangles, I stay inside of the homology class

Def. Homology group

Cycles

$$H_k(X) = \frac{\ker \partial_k}{\operatorname{im} \partial_{k+1}}$$

Boundaries

Example: What are the elements of the homology group?

First, we **find a cycle:**

$$c = (0,1) + (1,2) + (2,3) - (0,3) \in \ker \partial_1$$

$$\partial_1 (c) = (1) - (0) + (2) - (1) + (3) - (2) - (3) + (0) = 0$$

An **element of** $H_1(X)$ is given by the whole subset:

$$[c] := c + \operatorname{im} \partial_2 = \{c + \partial_2 \beta \ : \ \beta \in C_2(X)\}$$ *Homology class*

For instance, an **element of this subset** is:

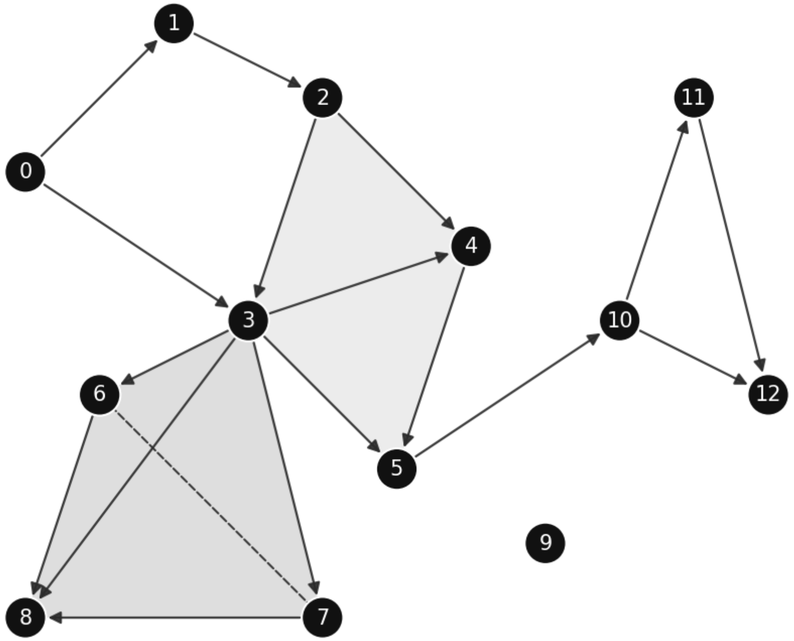
$$c + b = (0,1) + (1,2) + (2,4) - (3,4) - (0,3)$$

$$b = -(2,3) - (3,4) + (2,4) = \partial_2[-(2,3,4)] \in \operatorname{im} \partial_2$$

We call *homology representatives* to the elements of the *homology class*

Recap: simplicial homology

Betti numbers



Def. Betti number

$$\beta_k(X) = \dim H_k(X)$$

"Number of k-dimensional holes in X"

- $k = 0$: number of connected components
- $k = 1$: number of loops
- $k = 2$: number of cavities or bubles

Try it: What are the Betti numbers of the complex on the left?

Motivation: what we want to solve today

Topological inference problem

Suppose you are given the following point cloud, how can you design a test to conclude that this sample comes from a distribution over *two circles*?





Session 6 — Persistent Homology

Outline

- Induced maps in homology
- How to build simplicial complexes from point clouds
- Persistent Homology
  - Filtrations
  - Persistence modules
  - Barcodes

Induced maps in homology

Session 6 — Persistent Homology

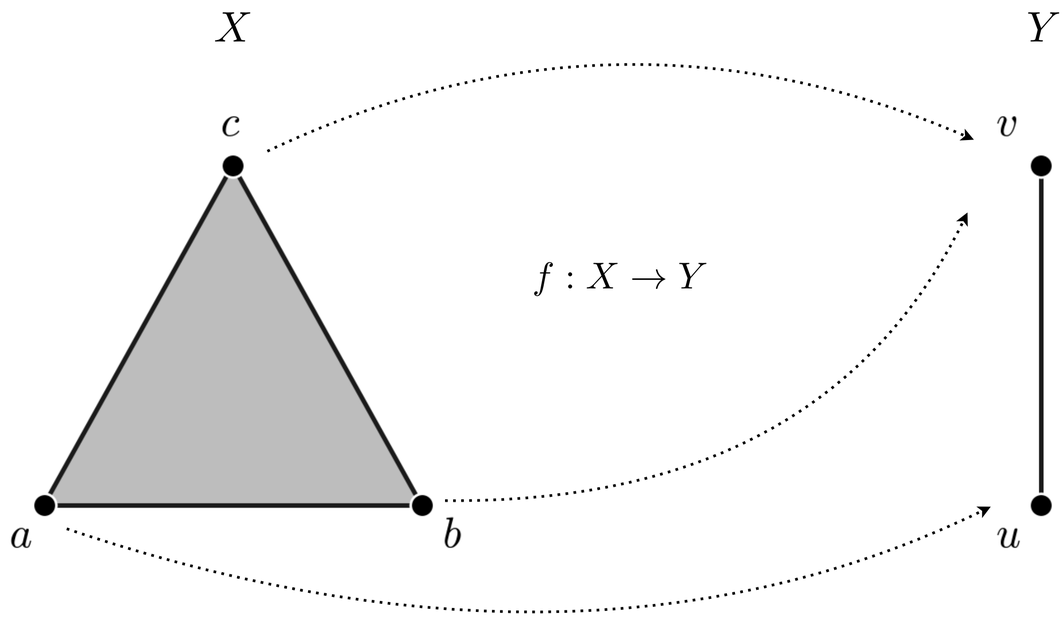
Simplicial maps

Maps between simplicial complexes

Def. Simplicial map

A simplicial map $f : X \to Y$ is a *map between the sets of vertices* of the simplicial complexes such that for every simplex $\sigma = \{x_0, \ldots, x_k\} \in X$, its image $f(\sigma) = \{f(x_0), \ldots, f(x_k)\} \in Y$ is also a simplex

Example:



$$f(a) = u, \quad f(b) = v, \quad f(c) = v$$

$$f(\{a, b\}) = \{u, v\}, \quad f(\{b, c\}) = \{v\}, \quad f(\{a, c\}) = \{u, v\}$$

$$f(\{a, b, c\}) = \{u, v\}$$

Induced maps in the chain space

Definition of induced maps at the chain level

Def. Induced map at the chain level

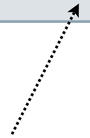
A simplicial map $f : X \to Y$ induces a linear map $f_\# : C_k(X) \to C_k(Y)$ for all $k \geq 0$ defined in the following way:

- We can write $\alpha \in C_k(X)$ in the basis of k-simplices as:

$$\alpha = \sum_{\sigma \in X_k} a_\sigma \cdot \sigma, \qquad a_\sigma \in \mathbb{R}$$

- Applying the simplicial map at each simplex, we can define the following chain in Y:

$$f_\#(\alpha) = \sum_{\sigma \in X_k \,:\, f(\sigma) \in Y_k} a_\sigma \cdot f(\sigma) \quad \in C_k(Y)$$



Notice that we require the image to be a k-simplex too!

Induced maps in the chain space

Induced maps commute with boundary operators

Important property

The induced map $f_\#$ *commutes* with the boundary maps $\partial_k^X : C_k(X) \to C_{k-1}(X)$ and $\partial_k^Y : C_k(Y) \to C_{k-1}(Y)$, that is:

$$f_\#\big(\partial_k^X(\alpha)\big) = \partial_k^Y\big(f_\#(\alpha)\big)$$

Proof.

For a k-simplex $\sigma = \{x_0, \ldots, x_k\} \in X$ we have two options:

- $f(\sigma) = (y_0, \ldots, y_k) \in Y_k$, that is, all vertices in σ are mapped to different vertices $f(x_i) = y_i \neq y_j = f(x_j)$, for $i \neq j$, and $f(\sigma)$ is a k-simplex of Y.
  In that case:

$$f_\#\big(\partial_k^X(\sigma)\big) = f_\#\left(\sum_{i=0}^{k}(-1)^i(x_0, \ldots, \widehat{x_i}, \ldots, x_k)\right) = \sum_{i=0}^{k}(-1)^i f(x_0, \ldots, \widehat{x_i}, \ldots, x_k) = \sum_{i=0}^{k}(-1)^i(y_0, \ldots, \widehat{y_i}, \ldots, y_k) = \partial_k^Y\big(f_\#(\sigma)\big)$$

Induced maps in the chain space

Induced maps commute with boundary operators

Important property

The induced map $f_\#$ *commutes* with the boundary maps $\partial_k^X : C_k(X) \to C_{k-1}(X)$ and $\partial_k^Y : C_k(Y) \to C_{k-1}(Y)$, that is:

$$f_\#\big(\partial_k^X(\alpha)\big) = \partial_k^Y\big(f_\#(\alpha)\big)$$

- Two vertices in σ are mapped to the same vertex of Y, $f(x_i) = y_i = y_j = f(x_j)$ for $i < j$. In this case $f_\#(\sigma) = 0$ because $f(\sigma) \notin Y_k$, and $\partial_k^Y\big(f_\#(\sigma)\big) = 0$.

  We now check $f_\#(\partial_k^X(\sigma)) = 0$ too:

$$f_\#\big(\partial_k^X(\sigma)\big) = \sum_{l=0}^{k} (-1)^l (y_0, \ldots, \widehat{y_l}, \ldots, y_k)$$

  Where the sum on the right only includes $(y_0, \ldots, \widehat{y_l}, \ldots, y_k) \in Y_{k-1}$

  - If $l \notin \{i, j\}$, then $(y_0, \ldots, \widehat{y_l}, \ldots, y_k)$ has at most $k-2$ distinct vertices, and it is not in Y_{k-1}, so all these terms do not appear in the sum
  - We only have left $(-1)^i (y_0, \ldots, \widehat{y_i}, \ldots, y_k) + (-1)^j (y_0, \ldots, \widehat{y_j}, \ldots, y_k) = (-1)^i (y_0, \ldots, \widehat{y_i}, \ldots, y_k) + (-1)^j (-1)^{j-i-1} (y_0, \ldots, \widehat{y_i}, \ldots, y_k) = 0$, where in the second equality we have moved y_i to the j-th position, changing the orientation accordingly.

- If more vertices in σ are mapped to the same vertex in Y, we apply a similar argument to conclude that both sides of the equality are 0 too.

Induced maps in the homology space

Induced maps

Def. Induced map at the homology level

A simplicial map $f : X \to Y$ induces a linear map $f_* : H_k(X) \to H_k(Y)$ for all $k \geq 0$ defined in the following way:

- An element $[\alpha] \in H_k(X)$ is a whole set $[\alpha] = \{\alpha + \partial_{k+1}\beta \,:\, \beta \in C_{k+1}(X)\}$ where, also, $\alpha \in \ker \partial_k^X \iff \partial_k^X \alpha = 0$
- We define $f_*([\alpha]) = \left[f_\#(\alpha)\right] = \left\{f_\#(\alpha) + \partial_k^Y \beta \,:\, \beta \in C_{k+1}(Y)\right\}$

We need to check that this map is *well-defined*:

- First, we check that $f_\#(\alpha)$ is a cycle:

$$\partial_k^Y\big(f_\#(\alpha)\big) = f_\#\big(\partial_k^X(\alpha)\big) = f_\#(0) = 0$$

- Now we check that two elements of the same class $[\alpha_1] = [\alpha_2] \in H_k(X)$ go to the same class in $H_k(Y)$. Two elements in the same class are related as: $\alpha_1 = \alpha_2 + \partial_k^X \beta, \quad \beta \in C_{k+1}(X)$.
  Then: $f_\#(\alpha_1) = f_\#(\alpha_2) + f_\#(\partial_k^X \beta) = f_\#(\alpha_2) + \partial_k^Y(f_\#(\beta))$ and thus

$$f_*([\alpha_1]) = [f_\#(\alpha_1)] = [f_\#(\alpha_2)] = f_*([\alpha_2])$$

Induced maps in the homology space

Two important properties

Properties of the induced maps

The induced homology map $f_* : H_k(X) \to H_k(Y)$ for all $k \geq 0$ satisfies the following two properties:

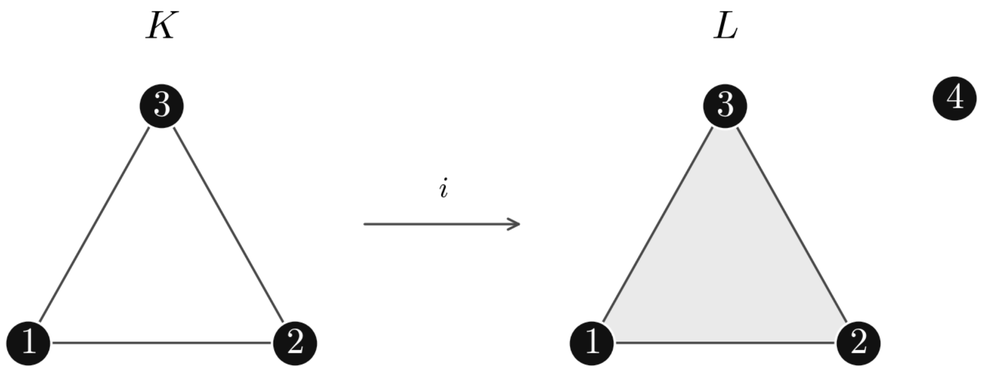
- If $\mathrm{id} : X \to X$ is the identity map, sending each vertex in X to itself, then $\mathrm{id}_* : H_k(X) \to H_k(X)$ is the identity map of vector spaces, which leaves every vector unchanged
- If $f : X \to Y$ and $g : Y \to Z$ then

$$(g \circ f)_* = g_* \circ f_*$$

An example for you to work on

Induced maps

Describe the induced maps for the 0, 1 and 2-dimensional homology groups of the following inclusion of simplicial complexes

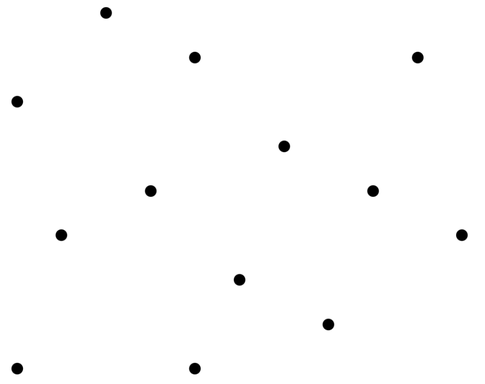


How to build simplicial complexes from point clouds

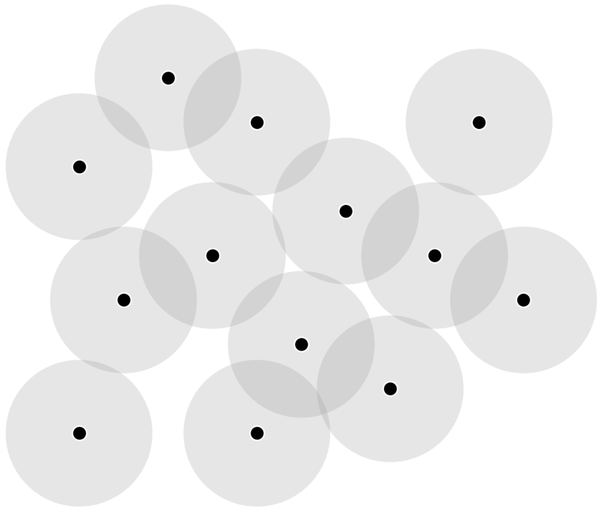
Session 6 — Persistent Homology

Point clouds

Our starting point



Point cloud



Point cloud + closed balls

A point cloud is a finite metric space (S, d) where:

- $S = \{s_1, \ldots, s_n\}$ is a **discrete set of points** (the vertices of the abstract simplicial complex we will build)
- $d : S \times S \to \mathbb{R}$ is a **distance function on the set**, which we can represent as the matrix of pairwise distances

$$\begin{pmatrix} d(s_1,s_1) & d(s_1,s_2) & \ldots & d(s_1,s_n) \\ d(s_2,s_1) & d(s_2,s_2) & \ldots & d(s_2,s_n) \\ \vdots & \vdots & \ddots & \vdots \\ d(s_n,s_1) & d(s_n,s_2) & \ldots & d(s_n,s_n) \end{pmatrix}$$

If we consider these points as living in a Euclidean space: $S \subset \mathbb{R}^D$, we can consider the **closed balls**:

$$B(x,r) := \{y \in \mathbb{R}^D : d(x,y) \leq r\}$$ Ball centered at x with radius r

Simplicial complexes

Building simplicial complexes from point clouds

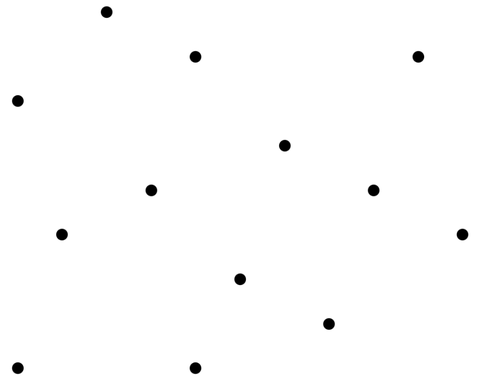
Finite metric space

Given (S, d) and $r \geq 0$, the **Čech complex** (or Nerve complex) at scale r is:

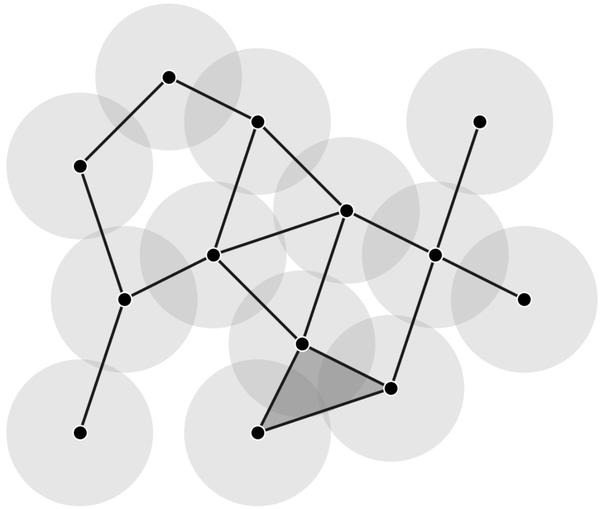
$$C_r(S, d) = \left\{ \{s_{i_0}, \ldots, s_{i_k}\} : \bigcap_{j=0}^{k} B(s_{i_j}, r) \neq \varnothing \right\}$$

Monotonicity

For $r \leq s$: $C_r(S) \subseteq C_s(S)$



Point cloud



Čech complex

Simplicial complexes

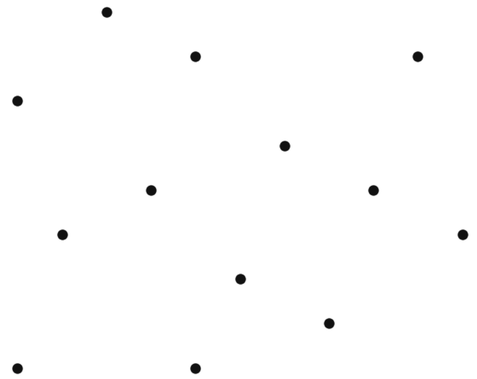
Building simplicial complexes from point clouds

Given (S, d) and $r \geq 0$, the **Vietoris-Rips complex** at scale r is:

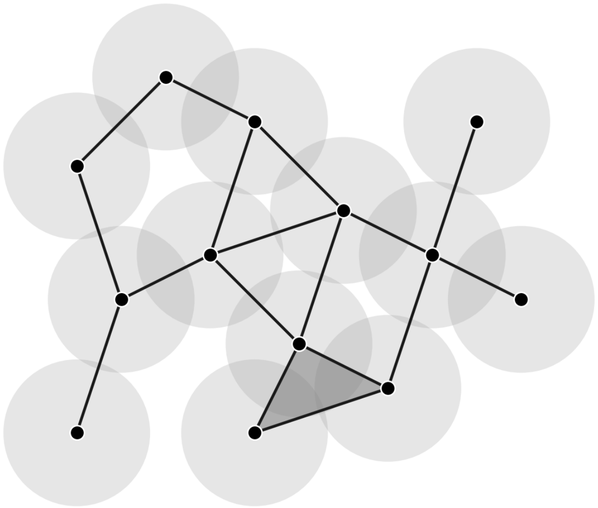
$$VR_r(S, d) = \left\{ \{s_{i_0}, \ldots, s_{i_k}\} : d(s_i, s_j) \leq 2r,\ \forall i_0 \leq i < j \leq i_k \right\}$$

Monotonicity

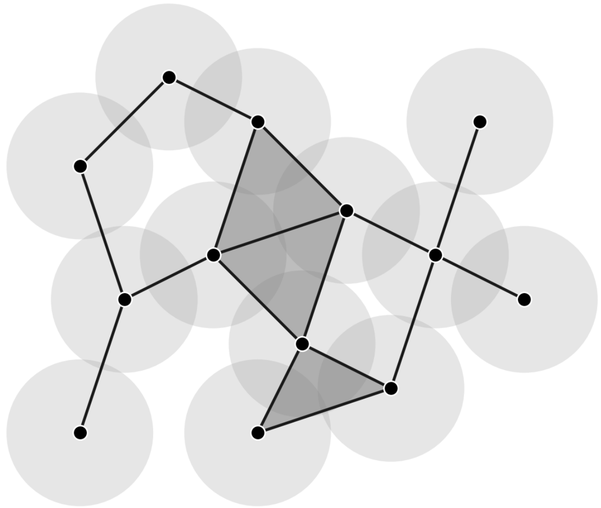
For $r \leq s$: $VR_r(S) \subseteq VR_s(S)$



Point cloud



Čech complex



VR complex

Simplicial complexes

Relating Čech and VR complexes

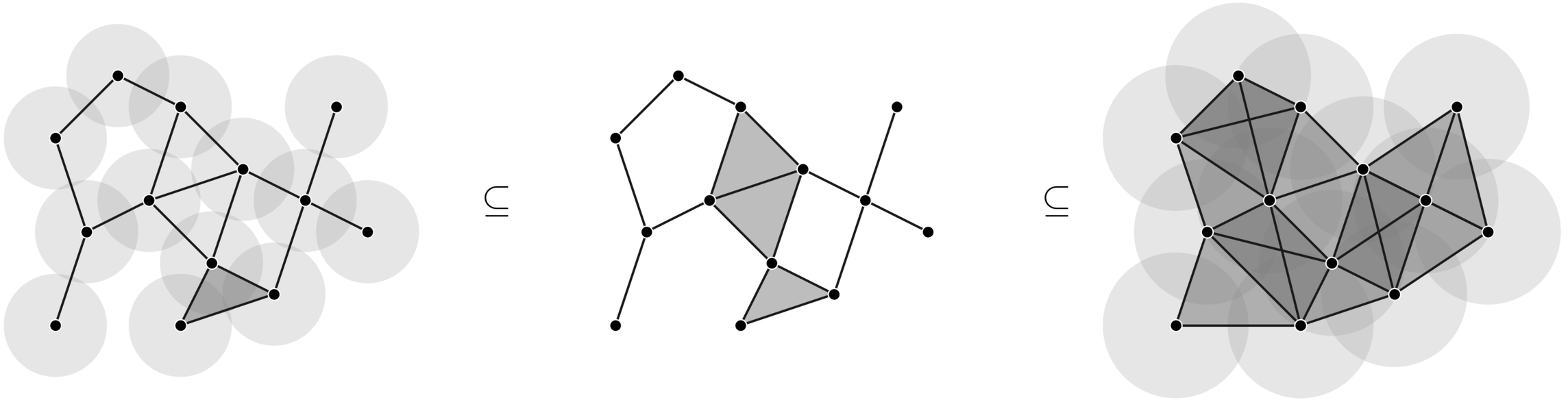
This containment is easy to prove

This one requires a bit more work

Lemma:

For $r \geq 0$: $\quad C_r(S) \subseteq VR_r(S) \subseteq C_{\sqrt{2}r}(S)$

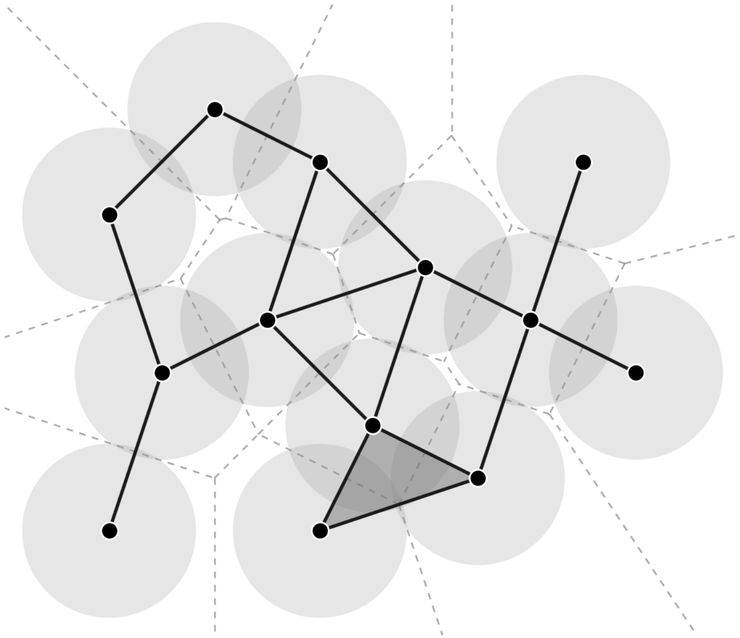
$C_r(S)$ $\quad VR_r(S)$ $\quad C_{\sqrt{2}r}(S)$



Simplicial complexes

Other simplicial complexes

- Alpha shapes and Alpha complexes
- Dowker complexes
- k-NN complexes



H. Edelsbrunner, D. Kirkpatrick and R. Seidel, “On the shape of a set of points in the plane", *IEEE Transactions on Information Theory* 29.4, 1983, pp. 551–559

C. H. Dowker, “Homology Groups of Relations”, *Annals of Mathematics* 56.1, 1952, pp. 84–95

M. Q. Le and D. Taylor, “Persistent Homology with k-nearest-neighbor Filtrations reveals Topological Convergence of PageRank”, 2022, arXiv: 2206.04725

To sum up

Today we have learnt how to...

- Obtain maps between homology groups from simplicial maps between simplicial complexes
- Build simplicial complexes from point clouds, approximating their shape at different scales

Thank you for your attention!



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Lecture 6 — Persistent Homology

Inés García Redondo — October 20, 2026